

Course code	Course Title		L	T	P	C
BCSE336L	Financial Data Analytics		2	0	0	2
Pre-requisite	NIL		Syllabus version			
			1.0			
Course Objectives						
1. To learn to model financial time series using liner ARMA type time series. 2. To study and analyze to test and model heteroscedastic effects using ARCH / GARCH type time series. 3. To learn how to test for unit root and construct ARMA models.						
Course Outcomes						
At the end of the course, the student will be able to 1. Approach and analyze any financial data. 2. Differentiate between various time series models. 3. Perform cross-validation of various financial models developed. 4. Forecast future observations on financial data.						
Module:1	Financial data and their properties		4 hours			
Asset Returns – Bond Yields and Prices – Implied Volatility – Examples and Visualization of financial data – Multivariate returns.						
Module:2	Linear models for financial time series		4 hours			
Simple autoregressive models – Simple moving average models – Simple ARMA models – Unit Root nonstationarity – Exponential smoothing.						
Module:3	Seasonal and Long memory models		4 hours			
Seasonal models – Regression models with time series errors – Long memory models.						
Module:4	Asset Volatility and Volatility models		4 hours			
Characteristics of Volatility – Structure of a model – Testing for ARCH Effect – ARCH Model – GARCH Model – GARCH-M Model – Exponential Garch Model – Threshold GARCH model – Stochastic volatility model – alternative approaches.						
Module:5	Applications of Volatility Models		4 hours			
Garch Volatility Term structure – Option pricing and hedging – Time Varying Correlations and Betas – Minimum Variance Portfolios – Prediction.						
Module:6	High Frequency Financial Data		4 hours			
Nonsynchronous trading – Bid ask spread of trading prices – Empirical characteristics of trading data – Models for price changes.						
Module:7	Value at Risk		4 hours			
Risk measure and Coherence – Risk metrics –Extreme value approach to Value at Risk – Peak over thresholds.						
Module:8	Contemporary Issues		2 hours			
	Total Lecture hours:		30 hours			
Text Book(s)						
1.	Ruey S. Tsay An Introduction to Analysis of Financial Data with R, Wiley, 2013.					
Reference Books						
1.	Analysis of Financial Time Series, by Ruey S. Tsay, 3rd edition, Wiley Series in Probability and Statistics, 2010.					
2.	William G. Foote, Financial Engineering Analytics: A Practice Manual Using R, 2018.					
3.	Statistical Analysis of Time-Series Data in SPlus, by Ren'e Carmona, Springer, March 4, 2004.					
Mode of Evaluation: CAT / written assignment / Quiz / FAT / Project / Seminar						
Recommended by Board of Studies			12-05-2022			
Approved by Academic Council			No. 66	Date	16-06-2022	